

E-COMMERCE SALES DASHBOARD

Name: Diya Hinge

Student ID: AF05220984

Batch Code: ANP-D4853

Project Name: E-Commerce Sales Dashboard using Microsoft Power BI

Project Guide: Mrs Rajeshree Thete

Objective

The objective of the **E-Commerce Sales Dashboard** is to analyze online sales data and transform raw business data into meaningful insights using **Microsoft Power BI**. The dashboard enables users to monitor sales performance, customer behavior, product categories, regional sales, payment methods, delivery performance, and product returns through interactive visualizations and dynamic filters.

Exploratory Analysis

- Analyze Total Sales, Orders, Customers, and Quantity Sold.
- Study monthly sales performance.
- Compare sales across product categories.
- Analyze regional sales performance.
- Identify customer purchasing patterns.
- Study payment method preferences.
- Analyze average delivery time.
- Evaluate return rate across orders.

Data Visualization

The dashboard contains interactive visualizations including:

- KPI Cards
- Clustered Column Chart
- Line Chart
- Bar Chart
- Doughnut Chart
- Pie Chart
- Interactive Slicers

These visualizations provide users with an easy-to-understand overview of business performance and support better decision-making.

Performance Analysis

- Identify the highest revenue-generating product category.
- Compare sales across different regions.
- Analyze monthly sales growth.
- Evaluate customer purchasing behavior.
- Study payment method distribution.
- Analyze delivery performance.
- Measure product return percentage.
- Compare sales quantity across categories.

Dashboard Development

The dashboard was developed using **Microsoft Power BI** with the following steps:

- Import E-Commerce dataset from CSV/Excel.
- Clean and transform data using **Power Query Editor**.
- Remove duplicate records and handle missing values.
- Create relationships between tables (Data Modeling).
- Create calculated columns and measures using **DAX**.
- Design KPI Cards and interactive visualizations.
- Add slicers for Month, Category, Region, and Payment Method.
- Publish a professional interactive dashboard.

Problem Statement

E-Commerce businesses generate a large volume of sales transactions every day, making manual analysis difficult and time-consuming.

This project aims to transform raw sales data into meaningful business insights by:

1. Importing and cleaning E-Commerce sales data.
2. Removing duplicate and inconsistent records.
3. Analyzing customer behavior and sales performance.
4. Comparing regional and category-wise sales.
5. Developing an interactive Power BI dashboard.

6. Supporting business decision-making through visual analytics.

Data Description

Source

The dataset was downloaded from **Kaggle**.

Data Model

The dataset follows a structured tabular model where each row represents one customer order.

Important Attributes

Attribute	Description
Order ID	Unique order number
Order Date	Date of purchase
Customer ID	Customer identifier
Customer Name	Customer name
Category	Product category
Product	Product name
Quantity	Number of items sold
Sales	Sales amount
Region	Sales region
Payment Method	Customer payment mode
Delivery Days	Delivery duration
Return Status	Returned or Not Returned

Data Relationships

- Each order belongs to one customer.
- Every product belongs to one category.
- Every customer belongs to one region.
- Every order uses one payment method.
- Every order has one delivery record.

- Sales amount is associated with each order.

Approach

1. Data Import

Software Used

- Microsoft Power BI Desktop
- Power Query Editor
- DAX
- Data Modeling

Process

- Import E-Commerce dataset.
- Verify data types.
- Check column names.
- Load data into Power BI.

2. Data Cleaning

The following cleaning operations were performed:

- Remove duplicate records.
- Handle missing values.
- Correct data types.
- Standardize category names.
- Standardize region names.
- Remove extra spaces.
- Format date columns.
- Validate Quantity and Sales values.
- Prepare the dataset for visualization.

3. Data Analysis and Visualization

KPI Cards

- Total Sales
- Total Orders

- Total Customers
- Quantity Sold
- Average Delivery Time
- Return Rate

Charts

- Sales by Category
- Monthly Sales Trend
- Sales by Region
- Orders by Payment Method
- Customer Age Distribution
- Return Status

Interactive Slicers

- Month
- Category
- Region
- Payment Method

Project Result

1. Data Collection and Preparation

The E-Commerce dataset was imported from Kaggle and cleaned using **Power Query** before analysis.

2. KPI Dashboard

Displays:

- Total Sales
- Total Orders
- Total Customers
- Quantity Sold
- Average Delivery Time
- Return Rate

Result:

Provides a quick summary of overall sales performance.

3. Category Analysis

Displays:

Sales by Category

Result:

Identifies the product categories generating the highest sales.

4. Regional Analysis

Displays:

Sales by Region

Result:

Shows which regions contribute the most to overall sales.

5. Monthly Sales Analysis

Displays:

Monthly Sales Trend

Result:

Helps identify seasonal trends and monthly business performance.

6. Payment Method Analysis

Displays:

Orders by Payment Method

Result:

Shows customers' preferred payment methods.

7. Customer Analysis

Displays:

Customer Age Distribution

Result:

Shows the distribution of customers across different age groups.

8. Return Analysis

Displays:

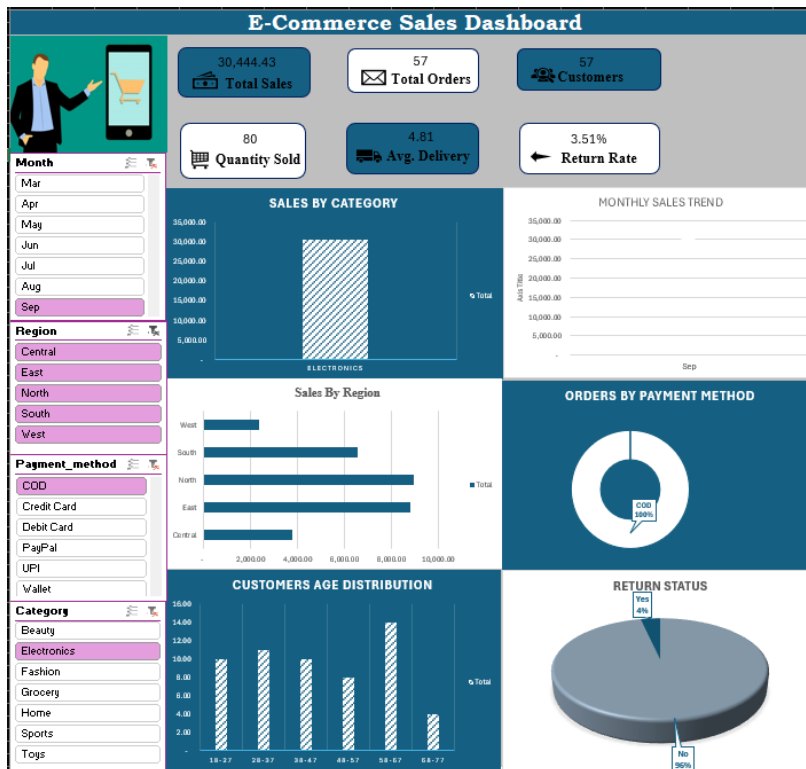
Return Status

Result:

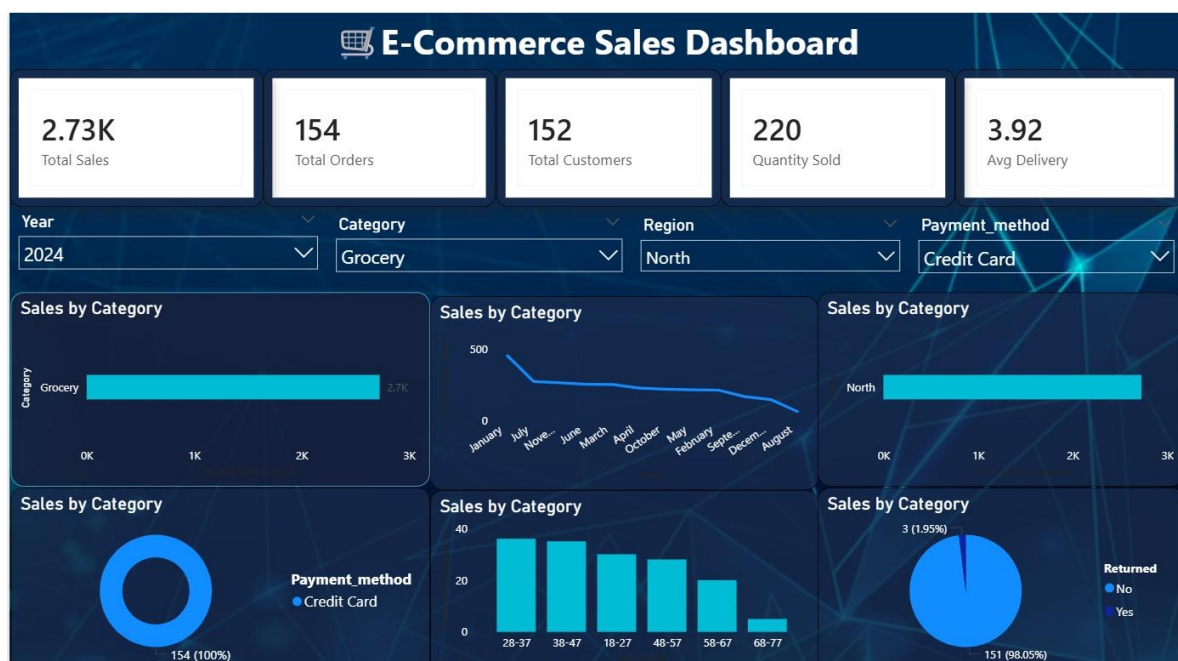
Shows the percentage of returned and non-returned orders.

Dashboard Screenshot :

Dashboard Using Excel :



Dashboard Using Power BI :



Conclusion

The **E-Commerce Sales Dashboard** successfully transforms raw sales data into meaningful business insights using **Microsoft Power BI**. The dashboard enables users to analyze sales performance, customer behavior, regional trends, payment methods, delivery performance, and product returns through interactive visualizations and dynamic filtering.

This project demonstrates the practical application of **Power Query, Data Modeling, DAX calculations, KPI creation, and interactive dashboard development** in Microsoft Power BI. Future enhancements may include profit analysis, sales forecasting, customer segmentation, inventory management, and real-time data integration.

This version closely matches the style and structure of your Healthcare Analytics Dashboard documentation while being fully adapted for your **Power BI E-Commerce Sales Dashboard**. It is suitable for college project submission or presentation.